

Mahir Puri

(672) 472-3123 | mahirpuri.17@gmail.com | mahirpuri.com | linkedin.com/in/mahir-puri | github.com/Mahir-Puri

EDUCATION

University of Victoria

Victoria, BC

Bachelor of Software Engineering – GPA: 4.0 / 4.0

Sept. 2024 – Dec. 2028

- **Relevant coursework:** Data Structures & Algorithms, Operating Systems, Distributed Systems, Database Systems, Object-Oriented Design, Statistics
- **Awards:** RBC Global Security Student Innovation Challenge (2026) | Certificate of Achievement, UVic Mathematics & Statistics (2025) | Scholarship for Academic Excellence (2024)

EXPERIENCE

Incoming Software Engineering Co-op

Sept. 2026 – Dec. 2026

Royal Bank of Canada (RBC) – Real-Time Payments Cloud Team

Toronto, ON

- Selected to return to RBC for a second co-op on the **Real-Time Payments Cloud Team**, with an incoming role developing and testing high-throughput **Java and Spring Boot** payment APIs deployed on **OpenShift** for RBC's Unified Global Payment Platform.

AI & Software Engineer Co-op

Jan. 2026 – Apr. 2026

Royal Bank of Canada (RBC) – Global Security, Cyber Technology Office

Toronto, ON

- Drove **69% faster remediation** across **500+ applications** by building **PySpark Medallion pipelines** on Azure Databricks unifying **5 security sources** with SCD Type 2, incremental loading, and data-quality gates.
- Cut pentest prep from **5–7 hours to under 5 minutes** by building an **agentic RAG system** (LangChain + LangGraph) with typed semantic schemas translating Gold-layer outputs into **LLM-ready context** across **20+ attack chains**.
- Provisioned **reproducible cloud infrastructure** across all pipelines using **Terraform and Docker** with least-privilege IAM policies – standardized deployments and eliminated environment inconsistencies across security teams.
- Eliminated **96+ hours of monthly manual work** by automating security reporting with **Python and SharePoint APIs**; idempotent execution guaranteed safe reruns across compliance-critical pipelines.
- Isolated a silent **SCD Type 2 phantom-update defect** by correlating Databricks job logs with the Delta transaction log – the bug had passed all pipeline validation while corrupting downstream AI context.
- Built **FastAPI and GraphQL** service layers over SQL materialized views – delivered an **Akamai API Security** view reused by two internal teams, exposing KPI/KRI metrics to **5+ teams** for SLA and remediation tracking.
- Won the **RBC's Global Security Student Innovation Challenge**, presenting Salus's architecture and impact to Directors, Senior Directors, and the **Chief Security Officer**.

PROJECTS

Strands – Multi-Agent Security Auditing Pipeline | *Python, FastAPI, Anthropic API, Docker* | [GitHub](#)

Ongoing

- Building a **compound AI system** with a planner, orchestrator, and three specialist sub-agents under least-privilege tool scoping; supervisor observes agent findings at runtime and revises the plan before the next agent executes.
- Implementing a **precision/recall/F1 evaluation harness** against benchmark repos with planted vulnerabilities, deterministic LLM cassette replay, and **50 automated tests** enforced through **GitHub Actions CI**.

RTPN – Real-Time Payments Network Simulator | *Java, Spring Boot, Kafka, PostgreSQL, Docker* | [GitHub](#)

2026

- Achieved **zero reconciliation inconsistencies** across all concurrent test runs by building a **double-entry RTGS ledger** in PostgreSQL with idempotency-keyed entries and a **Kafka** message rail with consumer retries and dead-letter handling.
- Validated ledger correctness under contention using a **Python simulator** that generated concurrent payment requests across independent institutions, with automated reconciliation checks covering retries, duplicate requests, and settlement failures.

Adversary – LLM Red-Teaming CI Harness | *Python, LLM-as-judge, GitHub Actions* | [GitHub](#)

2026

- Caught **100% of injected prompt attacks** in CI by building a **pip-installable harness** running **19 adversarial probes** across **6 attack categories**: injection, exfiltration, tool misuse, jailbreak, and system-prompt leak.
- Shipped a **severity-gated CI exit code** blocking merges on any failing probe; implemented an **injection-hardened LLM-as-judge** calibrated against human labels for auditable, consistent verdicts.

TECHNICAL SKILLS

Languages: Python, Java, SQL, Go, Bash

Backend & APIs: Spring Boot, FastAPI, GraphQL, REST APIs

Data & Databases: PySpark, Delta Live Tables, Medallion architecture, LangChain, LangGraph, PostgreSQL, DynamoDB, MySQL, Kafka

Cloud & Infrastructure: AWS (Lambda, S3, API Gateway), Azure Databricks, Terraform, Docker, Kubernetes, GitHub Actions

AI & Agentic: multi-agent orchestration, RAG pipelines, LLM evaluation, Anthropic API, Claude Code, prompt engineering